

In the 16th century, Polish astronomer and mathematician Nicolaus Copernicus challenged the prevailing view of the world by suggesting that the Sun, rather than the Earth, is at the center of the Universe.

MILESTONES

EARLY STUDIES

Studies astronomy and astrology at Kraków from 1491 to 1494; develops a passion for astronomy.

ADMINISTRATIVE POST

Accepts role as a canon at Frauenburg Cathedral in 1494; continues to study astronomy in his free time.

STELLAR REPUTATION

Receives an invitation from the Church in 1514 to advise on the reformation of the Julian calendar.

FIRST THEORY

Circulates a short text among friends in 1514 that outlines his theory of a heliocentric Universe.

FINAL PUBLICATION

Agrees to publish his controversial theory in 1543; dies as the book is printed and distributed.

Born in Poland in 1473, Nicolaus Copernicus studied astronomy and astrology in Kraków before traveling to Italy to study canon law (laws relating to the Church) at the University of Bologna. He later studied medicine at the University of Padua. After gaining a doctorate in canon law from the University of Ferrara in 1503, he moved to Wrocław, in western Poland, to devote his time to his greatest passion: astronomy. During these years, Copernicus formulated a cosmological theory that broke with centuries of tradition and eventually redefined the science of astronomy.

The established Universe

At the time Copernicus was working, there was a single accepted view of how the Universe was organized—the geocentric model. This model placed the Earth statically at the center of the Universe, with the planets, the Sun, the Moon, and the stars revolving around it. This view, proposed by Aristotle (see pp.14–17) and refined by Ptolemy (see p.33), had dominated cosmology for centuries. It was also entrenched within the doctrine of the Catholic Church, which considered humans to be the pinnacle of God's creation and the Earth to be at the center of everything. However, there were serious problems with the geocentric model. Chief among these was its failure to

Copernicus abandoned the geocentric model of the Universe after spending years observing the night sky. He used a device called a triquetrum to measure the altitudes of the stars and planets.

